IntelliTwin: Decision-Aware, Condition-Adaptive Conformal Calibration for Turbofan Remaining Useful Life Prognostics

Research synopsis — working draft; final results pending

Mayank Singh
IntelliTwin Research Initiative
Affiliation: pending institutional specification

Shiksha Pandey
IntelliTwin Research Initiative
Affiliation: pending institutional specification

Ruchi Gupta
IntelliTwin Research Initiative
Affiliation: pending institutional specification

Abstract—This research synopsis outlines the theoretical motivation, technical architecture, and confirmatory evaluation framework of the IntelliTwin initiative. The study investigates whether decision-aware, condition-adaptive conformal calibration can improve sequential maintenance utility while preserving reliable prediction-interval coverage for turbofan engine remaining useful life (RUL) estimation. We present the frozen experimental methodology, multi-regime dataset partitioning, base predictor matrix, exact rational engine-weighted quantiles, and 10,000-draw cluster-bootstrap evaluation protocol locked under Architecture Decision Record ADR-018 and verified under commit c163bb279e7427b8f1b6e75955af4969ff9370fd (v4 audit execution 550a7994d5746d3bcbfa28c26e7ed4f1a26a6153). Confirmatory empirical results on held-out test data remain pending experimental execution. Final author contributions, ordering, and institutional details require team agreement upon project completion.

Index Terms—Predictive maintenance, remaining useful life, conformal prediction, hierarchical inference, decision-aware calibration, digital twins, turbofan engines.

I. Introduction

Prognostics and Health Management (PHM) for high-value rotating machinery, such as commercial aviation turbofan engines, is critical to modern industrial operations [1], [2]. Operational maintenance strategies have traditionally navigated a severe trade-off between corrective maintenance and time-based preventive overhauls. Run-to-failure corrective strategies maximize the physical utilization of machine life but risk uncontained in-service failures, introducing severe flight delays and catastrophic safety hazards. Conversely, scheduled overhauls at fixed flight intervals prevent unexpected shutdowns but scrap substantial components with significant Remaining Useful Life (RUL) intact [3].

Predictive Maintenance (PdM) leverages multi-sensor telemetry to estimate RUL dynamically. Although modern deep neural networks and gradient boosting regressors achieve low point-prediction errors, point estimates $\hat{y}$ provide no measure of predictive uncertainty. In sequential replacement decisions, treating deterministic point estimates as ground truth is operationally hazardous: overestimating remaining life leads to in-service breakdown, while underestimating remaining life precipitates premature replacement [4], [5].

To enable mathematically reliable decision boundaries, prognostic models must produce calibrated prediction intervals $C(x) = [L(x), U(x)]$ targeting nominal coverage $1-\alpha$. Conformal prediction [6] provides a distribution-free framework for uncertainty quantification.

IntelliTwin uses an engine-weighted conformal construction motivated by hierarchical conformal prediction. Formal coverage results require the hierarchical sampling and exchangeability assumptions stated in the underlying theory. Equal engine weighting does not establish these assumptions for ordered, lifetime-stratified degradation trajectories. This study therefore evaluates marginal and subgroup coverage empirically. It does not claim guaranteed coverage for each engine, lifecycle stage, operating regime, complete trajectory, or adaptively selected maintenance time. Nominal 90

The IntelliTwin project investigates decision-aware, condition-adaptive conformal calibration on turbofan degradation benchmarks, seeking to align conditional prediction-interval reliability with sequential maintenance utility. The system is evaluated as an offline, trace-driven prognostic decision-support research prototype; physical engine integration and operational airline validation have not been demonstrated, and maintenance costs are expressed in Simulated Cost Units (SCU) representing synthetic benchmark preferences rather than validated airline economics.

II. Research Problem and Objectives

A. Research Problem Formulation

Existing literature separately evaluates conformal uncertainty calibration, operating-condition diagnostics, and uncertainty-aware maintenance scheduling. What remains insufficiently established is a Remaining Useful Life (RUL) prediction methodology that jointly adapts uncertainty calibration to operating conditions and downstream sequential replacement consequences while preserving reliable interval coverage.

Our study investigates whether decision-aware, condition-adaptive conformal calibration can improve sequential maintenance utility without sacrificing subgroup reliability or merely becoming more conservative.

B. Formal Research Questions and Hypotheses

In accordance with the frozen research protocol (ADR-018), four prespecified research questions are evaluated under version control prior to held-out evaluation:

- RQ1 (Sequential Maintenance Utility & Coverage): Can a decision-aware, condition-adaptive conformal RUL calibration method improve sequential maintenance utility while preserving reliable prediction-interval coverage? Evaluated on FD002 at 90% nominal coverage under primary_reference cost: requires cost improvement $\Delta_{\text{cost}} > 0.20$ SCU per engine over both global and condition-aware calibration, marginal, worst-regime, and worst-stage absolute coverage error ceilings < 0.05, and observed fleet failure rate increase $\Delta_{\text{fail}} \leq 0.0$.
- RQ2 (Conditional Subgroup Reliability): Can the proposed method improve reliability across operating regimes and degradation conditions without excessively widening prediction intervals? Evaluated across operating regimes (C_1–C_6) and degradation stages (Early/Mid/Late): requires worst-group conditional coverage error improvement $\Delta_{\text{error}} > 0.01$ over both baselines, absolute worst-group error < 0.05, and width and Winkler score deteriorations $D_{\text{width}}, D_{\text{score}} < 1.0$ cycle.
- RQ3 (Decision-Value Attribution): Are the maintenance benefits of the proposed method attributable to decision-aware uncertainty calibration rather than simply more conservative prediction intervals or earlier maintenance? Evaluated against four prespecified matched-conservatism controls across three families (matched width, widened interval, point safety margin, earlier intervention): requires cost improvement $\Delta_{\text{cost}} > 0.20$ SCU over every control, proposed-method coverage error ceilings < 0.05, and observed failure rate increase $\Delta_{\text{fail}} \leq 0.0$.
- RQ4 (Robustness & Generalization): Does the proposed method retain its calibration and maintenance-decision benefits across operating conditions, base RUL predictors, datasets, and repeated experimental runs? Evaluated across the balanced 6-cell matrix (FD001, FD002, FD004 × GBR, LSTM) weighted equally (1/6 each): requires weighted aggregate $\Delta_{\text{cost}} > 0.20$ SCU and $\Delta_{\text{error}} > 0.01$, cell-level non-harm ($\Delta_{\text{cost}} > -0.20$ SCU, $\Delta_{\text{error}} > -0.01$), and all cell-level coverage, width, score, and observed-failure safeguards.

III. Related Work and Positioning

A. Prognostic Models and Asymmetric Penalties

Prognostic modeling for aero-engines widely benchmarked on C-MAPSS [1] encompasses deep architectures including recurrent neural networks and gradient-boosted decision trees. While the C-MAPSS scoring metric penalizes overestimation ($d > 0$) more severely than underestimation ($d < 0$), base models are typically trained under symmetric loss functions.

B. Conformal Uncertainty Quantification

Conformal prediction [6] provides distribution-free marginal guarantees. Conformalized Quantile Regression (CQR) has been applied to aero-engines by Diao et al. [5] and Robinson [7]. However, standard split conformal methods assume observation exchangeability. In degradation trajectories, observations are clustered within engines; Lee, Barber, and Willett [8] established hierarchical conformal inference, placing $\frac{1}{m+1}$ probability mass at $+\infty$ to target valid coverage for unseen clusters under exchangeable group sampling.

IntelliTwin uses an engine-weighted conformal construction motivated by hierarchical conformal prediction. Formal coverage results require the hierarchical sampling and exchangeability assumptions stated in the underlying theory. Equal engine weighting does not establish these assumptions for ordered, lifetime-stratified degradation trajectories. This study therefore evaluates marginal and subgroup coverage empirically. It does not claim guaranteed coverage for each engine, lifecycle stage, operating regime, complete trajectory, or adaptively selected maintenance time. Nominal 90

C. Subgroup Diagnostics and Maintenance Utility

Auditing localized reliability under operating-regime shifts [9] and lifecycle degradation stages [10] demonstrates that marginal coverage frequently conceals localized failure in high-stress regimes. Concurrently, utility-directed conformal prediction [11] demonstrated the benefits of decision-aware nonconformity scores. IntelliTwin unites hierarchical engine-weighted conformal inference with asymmetric decision-aware score functions evaluated in sequential maintenance simulation.

IV. Proposed Methodology

A. Preprocessing and Predictive Feature Space

Evaluations span C-MAPSS FD001 (100 engines), FD002 (260 engines), and FD004 (249 engines). Splits are constructed strictly at the engine level: approx. 50% training, 20% calibration, 10% validation, and 20% held-out testing (Table I).

Preprocessing uses training-only z-score normalization. Means (μ_{train}) and population standard deviations (σ_{train}) are estimated with degrees of freedom ddof = 0. If a training standard deviation is exactly zero, it is replaced by an effective scale of 1.0, while the mean is still subtracted:

$$x_{\text{normalized}} = \frac{x - \mu_{\text{train}}}{s_{\text{train}}}, \quad s_{\text{train}} = \begin{cases} \sigma_{\text{train}} & \text{if } \sigma_{\text{train}} > 0, \\ 1.0 & \text{otherwise.} \end{cases} \tag{1}$$

For FD001, global normalization is fitted on training engines. For FD002 and FD004, operational settings are standardized, six operating regimes are learned via K-means ($k = 6, \text{random_state} = 2026, \text{n_init} = 10$) with lexicographic centroid ordering (C1–C6) and lower-index tie-breaking, and sensor normalization is fitted separately within training regimes. Preprocessing precedes window construction. The canonical 17-feature order is setting1, setting2, setting3, s2, s3, s4, s7, s8, s9, s11, s12, s13, s14, s15, s17, s20, s21.

Causal sliding windows of length $w = 30$ cycles with unit stride ($s = 1$) and no padding are extracted. GBR models use sklearn.ensemble.HistGradientBoostingRegressor on flattened 510-dimensional vectors; LSTM models use a single-layer unidirectional network (input = 17, hidden = 64, internal dropout = 0.0, followed by Dropout(0.20) and Linear(64, 1), trained via AdamW with learning rate 0.001, weight decay 10^{-4}, batch size 256, betas (0.9, 0.999), and $\epsilon = 10^{-8}$). Training targets use an upper cap of 125 cycles as a modeling convention. Point predictions are evaluated as $p = \max(0, \text{raw_prediction})$.

B. Hierarchical Engine-Weighted Conformal Calibration

To account for unequal engine trajectory lengths, the final hierarchical calibration distribution across m calibration engines satisfies:

$$F_{\text{final}}(q) = \frac{1}{m+1} \sum_{i=1}^{m} \left[\frac{1}{n_i} \sum_{t=1}^{n_i} \mathbf{1}_{\{\text{score}_{it} \le q\}} \right]. \tag{2}$$

Each finite endpoint carries mass $\frac{1}{(m+1)n_i}$. Total finite mass is $\frac{m}{m+1}$, and remaining mass $\frac{1}{m+1}$ is placed at $+\infty$. For regime calibration, required engine support is $m_g \ge \max(10, \lceil 1/\alpha \rceil - 1)$ (10 engines for 80% and 90%, 19 engines for 95%); otherwise execution falls back to the global threshold.

C. Decision-Aware Asymmetric Nonconformity Score

The proposed method (decision_aware_hcp) modulates scaling and asymmetry as a function of non-negative point prediction p:

$$\sigma(p) = \left(1 + \frac{p}{125}\right)^{\gamma}, \quad z(p) = \exp\left(-\frac{p}{20}\right) \tag{3}$$

$$a(p) = \sigma(p) \exp(\beta z(p)), \quad b(p) = \sigma(p) \exp(-\beta z(p)) \tag{4}$$

where $\gamma \in \{0, 1, 2\}$ and $\beta \in \{-\ln 2, 0, +\ln 2\}$. Nonconformity scores and inverted intervals are:

$$\text{score}(p, y) = \max\left(\frac{p - y}{a(p)}, \ \frac{y - p}{b(p)}, \ 0\right) \tag{5}$$

$$L = \max\left(0, \ p - q_g \cdot a(p)\right), \quad U = p + q_g \cdot b(p). \tag{6}$$

At fixed p, q and γ, negative β decreases $a(p)$ and increases $b(p)$. The lower and upper endpoints are therefore nondecreasing relative to $\beta = 0$. This does not establish earlier intervention or better maintenance utility because calibration thresholds are refitted and policy outcomes depend on the resulting trajectories. Validation candidate selection froze configuration gamma2_betam1 ($\gamma = 2, \beta = -\ln 2$) across all 6 cells under ADR-021.

D. Sequential Maintenance Simulator

A discrete-event simulation engine replays engine cycles from $t = 30$ to $T - 1$, executing the prespecified policy:

$$a_t = \begin{cases} \text{MAINTAIN_NOW}, & \text{if } L(t) \le 1.0 \text{ cycle}, \\ \text{CONTINUE}, & \text{otherwise.} \end{cases} \tag{7}$$

Preventive maintenance occurs immediately upon trigger, terminating the episode. Under the primary_reference scenario, preventive replacement incurs $C_{\text{PM}} = 1.0$ SCU, unplanned in-service failure at cycle T incurs $C_{\text{FAIL}} = 5.0$ SCU, and wasted life incurs $C_{\text{WASTE}} = 0.20$ SCU per unused cycle. Sensitivity scenarios evaluate $C_{\text{FAIL}} = 10.0$ and $C_{\text{WASTE}} = 0.40$. SCU values represent synthetic benchmark preferences rather than validated airline economics.

V. Experimental Evaluation Protocol

A. Statistical Uncertainty and Engine-Level Resampling

To account for trajectory autocorrelation, all comparative inferences use paired engine-cluster bootstrap resampling with replacement ($B = 10{,}000$ resamples) stratified by lifetime quartiles (Q_1–Q_4). The pseudo-random number generator is PCG64 seeded via SeedSequence([2026, 6, dataset_number]).

Multiplicity control across the four research questions is enforced at $\alpha_{\text{fam}} = 0.05/4 = 0.0125$ via Bonferroni correction as a prespecified nominal familywise adjustment, yielding 98.75% decision intervals (empirical quantiles 0.00625 and 0.99375). Percentile bootstrap inference is approximate; exact finite-sample error control is not claimed. The observed failure safeguard is an empirical

gate, not a confidence-interval proof of population safety. In accordance with intersection-union testing (IUT), full support for an RQ requires every single component condition to clear its boundary at the 98.75% decision level.

B. Frozen Decision Rules and Practical Margins

- RQ1 (Utility & Coverage): Lower CI of $\Delta_{\text{cost}} > 0.20$ SCU per engine vs. both global_hcp and regime_hcp; upper CI of marginal, worst-regime, and worst-stage absolute coverage errors < 0.05; observed fleet failure rate increase $\Delta_{\text{fail}} \leq 0.0$.
- RQ2 (Subgroup Reliability): Lower CI of $\Delta_{\text{error}} > 0.01$ vs. both baselines across operating regimes (C_1–C_6) and lifecycle stages (Early/Mid/Late); upper CI of worst-group error < 0.05; upper CI of MPIW and interval score expansion $D_{\text{width}}, D_{\text{score}} < 1.0$ cycle.
- RQ3 (Decision-Value Attribution): Lower CI of $\Delta_{\text{cost}} > 0.20$ SCU over all four prespecified matched-conservatism controls across three families (matched width, widened interval, point safety margin, earlier intervention) selected on validation engines under ADR-023; upper CI of proposed-method marginal, worst-regime, and worst-stage coverage errors < 0.05; observed fleet failure rate increase $\Delta_{\text{fail}} \leq 0.0$ against every control.
- RQ4 (Generalization): Weighted aggregate lower CI of $\Delta_{\text{cost}} > 0.20$ SCU and $\Delta_{\text{error}} > 0.01$, combined with cell-level non-harm bounds ($\Delta_{\text{cost}} > -0.20$ SCU, $\Delta_{\text{error}} > -0.01$) and cell-level coverage, width, score, and failure safeguards across all six mandatory evaluation cells.

VI. Current Project Status and Verification Milestones

A. Completed Preparatory Engineering & Governance

The research preparation and audit phases are complete and immutably frozen:

- Data Splits & Schema: Packaging verified under ADR-001; C-MAPSS dataset splits frozen across 609 engines in data/splits/; 17-feature schema locked under ADR-013.
- Base Predictors: Complete 30-model matrix trained across five official seeds (2026–2030) under ADR-019 and ADR-020; all 30 checkpoints verified on CPU in experiments/base_models/model_registry.json.
- Conformal Calibration: 90 calibrator bundles fitted and verified under ADR-021; configuration gamma2_betam1 frozen across all six cells in experiments/calibration/selection/selection_freeze.json.
- Pre-Test Verification Audit (v4): Under committed audit artifact experiments/pretest_verification_v4/pretest_audit.json, all verification gates ran cleanly with git_dirty = false. The pre-test readiness status reports PASS and PRETEST_READY. Reused historical evidence confirms 60 model/partition reproductions on validation and calibration partitions with an overall maximum absolute difference of 0.0.
- Confirmatory Protocol Freeze: Evaluation scripts, metrics, and matched controls are locked under experiments/pretest_verification_v4/protocol_freeze.json.

TABLE I
Dataset Partition Summary (ADR-011)

Dataset	Train (50%)	Cal (20%)	Val (10%)	Test (20%)
FD001	50	20	10	20
FD002	130	52	26	52
FD004	124	50	25	50
Total	304	122	61	122

B. Demarcation and Verified Status Statement

Base predictors and conformal calibrators are implemented and frozen. The committed v4 pre-test audit reports PASS and PRETEST_READY. Held-out execution has not started, and RQ1–RQ4 outcomes remain pending. Development validation and audit results do not establish confirmatory maintenance benefit, subgroup reliability, or robustness.

In accordance with prespecified research governance:

1) Held-Out Confirmatory Evaluation: Strictly NOT STARTED. Zero held-out test windows have been evaluated (pretest_audit.json).
2) RQ1–RQ4 Outcomes: Strictly PENDING. No performance numbers or hypothesis outcomes are reported from held-out data in this draft.
3) Incorporation Roadmap: Verified empirical findings will be incorporated following the prespecified 10-step protocol once held-out evaluation is executed.

VII. Limitations, Threats to Validity, and Conclusion

A. Research Limitations

1) Benchmark Simulation: Evaluations are conducted on synthetic NASA C-MAPSS data where degradation is monotonic. Operational aircraft engines experience maintenance reversibility and non-engine sensor faults not captured in the benchmark.
2) Synthetic Economic Scenarios: The Simulated Cost Units ($C_{\text{PM}} = 1.0, C_{\text{FAIL}} = 5.0, C_{\text{WASTE}} = 0.20$) represent study-defined benchmark preferences, not empirical airline accounting constants.
3) Conformal Theory Boundaries: IntelliTwin uses an engine-weighted conformal construction motivated by hierarchical conformal prediction. Formal coverage results require the hierarchical sampling and exchangeability assumptions stated in the underlying theory. Equal engine weighting does not establish these assumptions for ordered, lifetime-stratified

degradation trajectories. This study therefore evaluates marginal and subgroup coverage empirically. It does not claim guaranteed coverage for each engine, lifecycle stage, operating regime, complete trajectory, or adaptively selected maintenance time. Nominal 90

4) Research Prototype Boundaries: The system is an offline, trace-driven prognostic decision-support research prototype; physical engine integration and operational airline validation have not been demonstrated.

B. Current Status and Conclusion

Base predictors and conformal calibrators are implemented and frozen. The committed v4 pre-test audit reports PASS and PRETEST_READY. Held-out execution has not started, and RQ1–RQ4 outcomes remain pending. Development validation and audit results do not establish confirmatory maintenance benefit, subgroup reliability, or robustness.

The reviewed repository baseline is commit c163bb279e7427b8f1b6e75955af4969ff9370fd, with current committed readiness evidence in experiments/pretest_verification_v4/. All methodology, base models, calibrator bundles, and evaluation contracts are completely frozen and audited. Upon completion of the official held-out test evaluation, verified empirical results will be incorporated to resolve RQ1–RQ4.

References

[1] A. Saxena, K. Goebel, D. Simon, and N. Eklund, "Damage propagation modeling for aircraft engine run-to-failure simulation," in 2008 International Conference on Prognostics and Health Management. IEEE, 2008, pp. 1–9.

[2] Z. Huang, Y. Shen, J. Li, M. Fey, and C. Brecher, "A survey on AI-Driven digital twins in Industry 4.0: Smart manufacturing and advanced robotics," Sensors, vol. 21, no. 19, p. 6340, 2021.

[3] H. K. Shehadeh, "Evaluating the impact of transitioning maintenance strategy from reactive to proactive in power generation companies: An empirical analysis," WSEAS Transactions on Business and Economics, vol. 21, pp. 838–856, 2024.

[4] C. Chen, J. Shi, N. Lu, Z. H. Zhu, and B. Jiang, "Data-driven predictive maintenance strategy considering the uncertainty in remaining useful life prediction," Neurocomputing, vol. 494, pp. 79–88, 2022.

[5] R. Diao, M. Zhou, G. Meng, and S. Wang, "Turbofan engine remaining useful life prediction with reliable prediction intervals via LSTM-Based quantile regression and conformal calibration," Sensors, vol. 26, no. 7, p. 2249, 2026.

[6] A. Javanmardi and E. Hüllermeier, "Conformal prediction intervals for remaining useful lifetime estimation," International Journal of Prognostics and Health Management, vol. 14, no. 2, p. 3417, 2023.

[7] C. D. Robinson, "Remaining useful life estimation for aircraft engines with risk-aware prediction intervals via conformalized quantile regression," International Journal of Prognostics and Health Management, vol. 17, no. 1, p. 4724, 2026.

[8] Y. Lee, R. F. Barber, and R. Willett, "Distribution-free inference with hierarchical data," arXiv preprint arXiv:2306.06342v4, 2023, revised August 2025 (v4 inspected); original preprint 2023. [Online]. Available: https://arxiv.org/abs/2306.06342v4

[9] S. Yang, J. Wang, and Y. Wang, "Empirical calibration and conditional-reliability diagnostics for bearing RUL prediction under operating-regime shift," arXiv preprint arXiv:2607.08273v1, 2026. [Online]. Available: https://arxiv.org/abs/2607.08273

[10] A. BenAbdennour, "Leakage-controlled grouped validation and interval-safety calibration for RUL/SOH prediction using engine, battery, and bearing degradation data," Machines, vol. 14, no. 10, p. 1124, 2026.

[11] S. Cortes-Gomez, C. M. Patiño, Y. Byun, S. Wu, E. Horvitz, and B. Wilder, "Utility-directed conformal prediction: A decision-aware framework for actionable uncertainty quantification," in International Conference on Learning Representations (ICLR), 2025, primary proceedings: https://proceedings.iclr.cc/paper_files/paper/2025/hash/0c6b452f1bbfb6905f6bac957d73b321-Abstract-Conference.html. [Online]. Available: https://openreview.net/forum?id=iOMnn1hSBO